Spectral Optimization Across Vision Architecture Families: A Controlled Empirical Study

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Abstract

Optimizer performance in vision models depends critically on architecture: SGD collapses to below 12% accuracy on layer-normalized architectures while achieving 70% on batch-normalized ones, a gap that hyperparameter tuning cannot close. We conduct a controlled empirical study of spectral optimizers—Muon, SOAP, and Shampoo—which exploit weight-matrix spectral structure through Newton-Schulz iteration or Kronecker-factored preconditioning, evaluating them across three architecture families (ResNet-50, DeiT-S, ConvNeXt-T), three datasets, and two training horizons. The spectral advantage is a *family-level* property: on CIFAR-100—the only dataset where all three were evaluated—all outperform AdamW by +9 to +16 pp. Muon, tested across all three datasets, maintains gaps of +5.9 to +24.3 pp. This advantage is architecture-dependent—largest on layer-normalized networks—and persists across extended training schedules, though gap narrowing varies by architecture. Layer-wise analysis reveals distinct mechanisms: stable-rank divergence correlates positively with optimizer benefit on CNNs ($\rho = +0.45$) but negatively on transformers ($\rho = -0.40$). A normalization-swap experiment on ResNet-50 provides causal evidence that normalization type mediates the spectral advantage.

1 Introduction

Modern vision architectures have diverged rapidly in structural design: vision transformers use layer normalization and global self-attention Touvron et al. [2021], hybrid architectures combine depthwise separable convolutions with layer normalization Liu et al. [2022], while classical CNNs retain batch normalization He et al. [2016]. Yet optimizer selection has not adapted accordingly—practitioners still default to AdamW or SGD regardless of these structural differences, treating optimizer choice as secondary to architecture and data.

This neglect is consequential. SGD collapses to 11.51% accuracy on layer-normalized ConvNeXt-T while achieving 69.50% on batch-normalized ResNet-50, suggesting normalization type as a critical factor—a hypothesis we examine in Section 5.3. The same task remains accessible to other optimizers, revealing that the optimizer-architecture interaction is not a minor hyperparameter issue but a fundamental compatibility constraint that current practice ignores.

Why do some optimizers transfer across architectures while others fail on certain architectures? Existing first-order methods—AdamW with diagonal curvature estimates, SGD with raw gradient direction—do not explicitly account for how architectural design reshapes the spectral properties of weight gradients. Prior optimizer comparisons typically evaluate single architectures Schmidt et al.

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[2021] or lack controlled protocol isolation Choi et al. [2020], making it impossible to disentangle optimizer capability from recipe engineering.

We hypothesize that optimizers exploiting *weight-matrix spectral structure*—through Newton–Schulz orthogonalization or Kronecker-factored preconditioning Bernstein and Newhouse [2024]—should exhibit architecture-robust performance precisely because they adapt to the spectral geometry regardless of how normalization and layer composition reshape it. We test this through a controlled study of three spectral optimizers (Muon Jordan et al. [2024][1], SOAP Vyas et al. [2024], Shampoo Gupta et al. [2018]) against AdamW Loshchilov and Hutter [2019] and SGD, spanning three architecture families, three datasets, and two training horizons, with independently tuned learning rates per optimizer–architecture pair (Section 4.2).

Our experiments confirm the family-level hypothesis across all tested configurations, revealing architecture-dependent mechanisms.

While BOCB [Li et al., 2024] diagnoses coupling bias and Kunstner et al. explain why SGD fails on transformers [Kunstner et al., 2024], neither tests spectral methods nor identifies the architectural mediator. Our work bridges these lines: we evaluate the spectral family that Bernstein & Newhouse unified theoretically, and identify normalization—not architecture topology—as the mechanism that determines when spectral preconditioning helps.

Contributions.

- The first controlled empirical test of whether Bernstein & Newhouse's theoretical family Bernstein and Newhouse [2024] shares practical properties, revealing that normalization type—not architecture family—is the primary mediator of optimizer-architecture coupling.
- Causal evidence that the spectral advantage is mediated by normalization type: replacing BatchNorm with LayerNorm on ResNet-50 collapses AdamW by $\sim$17 pp while Muon is preserved (+1.64 pp), and layer-wise stable-rank-divergence correlations reverse sign between CNNs ($\rho{=}+0.45$) and transformers ($\rho{=}-0.40$).
- Practical guidance: spectral methods are architecture-robust candidates across the training conditions tested, achieving superior accuracy at substantially lower training cost.
- A corruption robustness evaluation on CIFAR-100-C showing that spectral optimizers, particularly Muon, improve not only clean accuracy but also robustness to common input corruptions (mCE 89–93% on three of four architecture configurations vs. AdamW baseline of 100%).

2 Related Work

Optimizer–architecture coupling. The observation that optimizer performance depends on architecture is well-established but poorly addressed. SGD dominated the convolutional era He et al. [2016] but fails on modern layer-normalized architectures Liu et al. [2022], Touvron et al. [2021], driving a wholesale shift to AdamW Loshchilov and Hutter [2019]. Large-scale benchmarks confirm that no single optimizer dominates across architectures Choi et al. [2020], Schmidt et al. [2021], Dahl et al. [2024], and Li et al. Li et al. [2024] formalize this as *backbone–optimizer coupling bias*, showing that optimizer rankings shift substantially with backbone choice. The underlying mechanism involves distinct optimization landscapes: batch normalization smooths loss surfaces in ways that layer normalization does not Santurkar et al. [2018], and Vision Transformers converge to sharper minima than CNNs Chen et al. [2022]. However, these studies diagnose the problem without offering a solution—they characterize *which* optimizer works where, but not *why* certain methods transfer and others fail. Our work addresses this gap by evaluating spectral optimizers, absent from prior benchmarks, and by providing layer-wise analysis that reveals where in the network coupling originates.

Spectral and preconditioned optimization. A parallel research thread has developed optimizers that exploit gradient matrix structure beyond diagonal curvature estimates. K-FAC Martens and Grosse [2015] pioneered Kronecker-factored natural gradient; Shampoo Gupta et al. [2018], Anil et al. [2021] extended this to full-matrix preconditioning with practical scalability [Shi et al., 2023]; SOAP Vyas et al. [2024] introduced data-driven eigenbasis adaptation; and Muon Jordan et al. [2024] takes a distinct approach via Newton–Schulz orthogonalization, performing steepest descent under

[1] At the time of writing, the Muon reference is a blog post and has not been peer-reviewed.

the spectral norm (Appendix F). Bernstein & Newhouse Bernstein and Newhouse [2024] unify these methods theoretically, proving that Adam, Shampoo, and Muon each correspond to steepest descent under different matrix norms, while Pethick et al. Pethick et al. [2025] formalize this further through norm-constrained optimization oracles. This unification motivates treating them as a *spectral optimizer family*—yet no prior work evaluates whether the family shares empirical properties (such as architecture robustness) beyond theoretical structure. Our study provides this evaluation across diverse vision architectures.

Complementary optimizer directions. Other recent optimizers target computational cost or tuning difficulty—Lion Chen et al. [2023] via program search, Sophia Liu et al. [2024] via diagonal Hessian clipping, Prodigy Mishchenko and Defazio [2024] via learning-rate-free estimation, GaLore Zhao et al. [2024] via low-rank projection—rather than architecture robustness [Defazio et al., 2024]. Our focus is complementary: which structural property determines whether an optimizer transfers across architectural designs.

3 Spectral Analysis Framework

3.1 Spectral Optimization via Orthogonalized Updates

Muon Jordan et al. [2024], Liu and Su [2025] optimizes 2D weight matrices by orthogonalizing the gradient before applying the update, effectively performing a Newton-like step in the spectral domain. Given a gradient matrix $G \in \mathbb{R}^{m\times n}$ for a weight matrix W, the goal is to compute an approximation to $(GG^\top)^{-1/2}G$, which projects the gradient onto the Stiefel manifold of orthonormal row matrices. Rather than forming this matrix inverse explicitly, Muon uses Newton–Schulz iteration to approximate it efficiently. Five iterations of a quintic Newton–Schulz polynomial orthogonalize G (starting from $X_0 = G/\|G\|_F$; iteration formula, coefficients, and convergence analysis in Appendix F). The orthogonalized gradient serves as the update direction, combined with Nesterov momentum ($\beta = 0.95$).

This procedure captures inter-parameter curvature structure that first-order methods miss. SGD updates along the raw gradient direction; AdamW applies element-wise scaling via running second-moment estimates, which amounts to a diagonal approximation to the curvature. Muon's orthogonalization operates on the full gradient matrix, providing a non-diagonal second-order approximation related to K-FAC Martens and Grosse [2015], but without explicitly maintaining Kronecker-factored estimates. As described in Section 4, Muon is applied exclusively to 2D weight matrices (convolution kernels reshaped to 2D and linear layers), while remaining parameters are handled by a separate AdamW instance (Appendix Table 5 details the parameter routing).

3.2 Layer-wise Spectral Analysis Framework

Global accuracy differences between optimizers do not reveal *where* in the network one optimizer outperforms another. We introduce two complementary analysis tools that localize optimizer effects to individual layers.

Weight Swap Protocol. Given a Muon-trained model with weights $\theta^{\text{Muon}} = \{W_l^M\}$ and an AdamW-trained model with weights $\theta^{\text{AdamW}} = \{W_l^A\}$, we construct a hybrid model $\theta^{\text{Muon}\setminus l\leftarrow\text{AdamW}_l}$ by replacing the weights of a single layer l in the Muon model with the corresponding AdamW weights, keeping all other layers at their Muon-trained values. We define the per-layer Muon benefit:

$$\Delta a_l = \text{acc}(\theta^{\text{Muon}}) - \text{acc}(\theta^{\text{Muon}\setminus l\leftarrow\text{AdamW}_l}), \tag{1}$$

where $\text{acc}(\cdot)$ denotes validation accuracy. $\Delta a_l > 0$ indicates that Muon's training at layer l outperforms AdamW; $\Delta a_l < 0$ indicates the reverse. The beneficial layer ratio, the fraction of layers where $\Delta a_l > 0$, aggregates this across all eligible layers.

Spectral Correlation Analysis. To connect the weight swap results with spectral structure, we measure how each optimizer shapes the *stable rank* of each layer's weight matrix:

$$\text{sr}(W_l) = \frac{\|W_l\|_F^2}{\sigma_1(W_l)^2}, \tag{2}$$

where $\sigma_1(W_l)$ is the largest singular value. Stable rank captures the effective dimensionality of the weight matrix: higher values indicate a more uniform singular value distribution, meaning the matrix spreads its representational capacity across more directions. We choose stable rank over alternatives (effective rank, condition number, spectral gap) because it is bounded, well-defined for rank-deficient matrices, and permits direct cross-architecture comparison without architecture-specific normalization. For convolutional layers, we reshape the 4D kernel tensor to a 2D matrix before computing $\mathrm{sr}(W_l)$. We compute the per-layer stable rank divergence $\Delta\mathrm{sr}_l = \mathrm{sr}(W_l^{\text{Muon}}) - \mathrm{sr}(W_l^{\text{AdamW}})$: positive values indicate Muon spreads spectral energy more uniformly (higher effective rank), negative values indicate Muon concentrates energy into fewer directions. We evaluate the Spearman rank correlation between $\Delta\mathrm{sr}_l$ and Δa_l (Eq. 1); results are reported in Section 5.3.

3.3 Evaluation Metrics and Statistical Protocol

We report best validation accuracy (early-stopping checkpoint) throughout, in percentage points (pp), to decouple optimizer comparison from convergence speed. Key experiments use 3–4 seeds (from $\{42, 0, 1, 2\}$; Appendix Table 11), reporting mean $\pm$ std (ddof $= 1$). Independent 50-epoch runs with cosine annealing ($T_{\max} = 50$) and the same per-pair learning rates test gap persistence without schedule-length confounding.

4 Experimental Setup

4.1 Datasets and Architectures

We evaluate on three benchmarks: CIFAR-100 Krizhevsky and Hinton [2009] (60K images, 100 classes, 32×32 resized to 224×224), Tiny-ImageNet Le and Yang [2015] (100K images, 200 classes, 64×64), and ImageNet-100 Deng et al. [2009] (a 100-class subset at native 224×224). Training augmentation uses `Resize(256)`, `RandomCrop(224)`, and random horizontal flip.

We select three architectures with comparable parameter counts (22–29M) spanning the major design paradigms: ResNet-50 He et al. [2016] (batch normalization, $\sim$25.6M), DeiT-S Touvron et al. [2021] (layer normalization, multi-head attention, $\sim$22M), and ConvNeXt-T Liu et al. [2022] (layer normalization, depthwise separable convolutions, $\sim$28.6M). All models are trained from scratch with He initialization He et al. [2015] using `timm` Wightman [2019] without pretrained weights.

4.2 Optimization Protocol

We compare five optimizers: Muon Jordan et al. [2024] (spectral-norm orthogonalization via Newton–Schulz iteration, Nesterov β=0.95, applied to 2D weight matrices with a companion AdamW for remaining parameters), SOAP Vyas et al. [2024] (Kronecker-factored eigenbasis preconditioning), Shampoo Gupta et al. [2018] (full-matrix preconditioning)[2], AdamW Loshchilov and Hutter [2019], Kingma and Ba [2015] (β_1=0.9, β_2=0.999, wd 0.05), and SGD Sutskever et al. [2013] (Nesterov momentum 0.9, wd 10^{-4}). All configurations use batch size 128, cosine annealing, and 20 epochs for the main comparison with 50-epoch extensions for convergence analysis.

To ensure fair comparison, each optimizer–architecture–dataset combination receives an independent learning rate selection:

(a) **Sweep:** A 5-epoch sweep over 3–7 log-spaced rates Schmidt et al. [2021]; 18 of 27 configurations confirmed a clear peak (accuracy declining at adjacent rates).
(b) **Special cases:** 7 monotonic configurations (all spectral optimizers) were validated via 20-epoch saturation runs confirming no further gain; 2 used literature defaults (Muon on R50 CIFAR-100, lr=0.02; SGD on R50, lr=0.1).

The 7/9 monotonic rate (all spectral) suggests 20-epoch gaps are a conservative proxy, since optimal learning rates may lie beyond the swept range (Appendix Tables 7–9). Selected rates and full configurations are in Appendix A.

[2]SOAP and Shampoo use the `heavyball` library implementation: `https://github.com/ClashLuke/HeavyBall`.

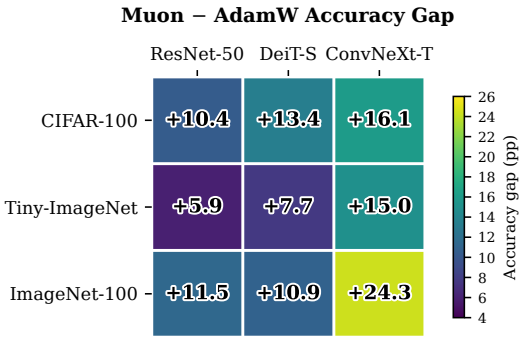


Figure 1: Spectral–AdamW accuracy gap (pp) at 20 epochs. ConvNeXt-T consistently exhibits the largest gap (+15.0–+24.3 pp); all gaps are positive across the full 3×3 matrix.

4.3 Evaluation and Reproducibility

We report best validation accuracy across all epochs (i.e., the early-stopping checkpoint) and quantify optimizer gaps in percentage points (pp). Key experiments are replicated with 3 seeds, reporting mean $\pm$ std. Independent 50-epoch runs verify whether gaps persist at longer horizons. All results are obtained on a single NVIDIA RTX 5090 GPU with PyTorch and `timm`; code and training logs will be released upon publication.

5 Results

We organize results around five questions: (1) whether the spectral advantage is family-level and how it depends on architecture (§5.1), (2) whether it persists with longer training (§5.2), (3) what layer-wise mechanisms underlie it (§5.3), (4) whether it generalizes to modern training recipes (§5.4), and (5) whether the advantage is attributable to spectral preconditioning itself (§5.5).

5.1 Spectral Advantage: Magnitude and Architecture Dependence

If spectral optimizers succeed by exploiting weight-matrix structure, their advantage should be universal across architectures but variable in magnitude. Table 1 confirms this prediction: Muon outperforms AdamW across all nine architecture–dataset pairs, with gaps ranging from $+5.9$ pp (ResNet-50/Tiny-ImageNet) to $+24.3$ pp (ConvNeXt-T/ImageNet-100), as visualized in Figure 1. Multi-seed variance is low (0.10–2.17 pp; Appendix Table 11), confirming that observed gaps substantially exceed run-to-run noise.

Spectral Family Comparison on CIFAR-100. A critical question for interpretation is whether the advantage is specific to Muon's Newton–Schulz mechanism or shared across spectral methods. We evaluate SOAP Vyas et al. [2024] and Shampoo Gupta et al. [2018] on CIFAR-100 under the same protocol (Table 1; full comparison in Appendix Table 18). SOAP matches Muon within noise on ResNet-50 and ConvNeXt-T and outperforms it on DeiT-S ($+3.10$ pp, $p < 0.001$). Within the spectral family, Shampoo leads on ResNet-50 ($+1.37$ pp over Muon), SOAP on DeiT-S ($+3.10$ pp, $p<0.001$), and Muon on ConvNeXt-T ($+4.05$ pp)—each architecture favoring a different spectral variant. All spectral optimizers outperform AdamW by $+9$–$+16$ pp on CIFAR-100, with intra-family differences (1.4–4.1 pp) approximately 4–8$\times$ smaller than the spectral–AdamW gap, establishing the benefit as a *family-level* property (Appendix A).

Architecture dependence. Figure 1 reveals a consistent ordering: ConvNeXt-T exhibits the largest gap at every dataset ($+16.08$, $+15.00$, $+24.30$ pp), 1.2–2.5$\times$ larger than the other architectures, while ResNet-50 shows the smallest. ConvNeXt-T consistently shows the largest gap while ResNet-50 shows the smallest, a pattern consistent across all three spectral methods (Appendix Table 18). We discuss the mechanistic basis for this ordering in Section 6.

Table 1: Best validation accuracy (%) at 20 epochs. Multi-seed results (± std) where available; *single-seed (seed = 42). Bold = best per column. Gap = best spectral − AdamW in pp. †Muon−AdamW only (SOAP/Shampoo evaluated on CIFAR-100 only).

Dataset	Optimizer	ResNet-50	DeiT-S	ConvNeXt-T
CIFAR-100	Shampoo	**78.25**±0.18	63.53±0.05	70.57±0.29
	SOAP	77.28±0.20	**65.77**±0.39	74.40±0.43
	Muon	76.88±0.27	62.67±0.41	**74.62**±0.15
	AdamW	67.85±0.17	52.33±0.34	58.54±2.17
	SGD	69.50±0.39	43.98±0.48	11.51*
	Gap	**+10.40**	**+13.44**	**+16.08**
Tiny-ImageNet	Muon	**62.66**±0.32	**48.02**±0.21	**58.92**±0.19
	AdamW	56.76±0.45	40.33±0.42	43.92±0.29
	SGD	60.04*	34.22*	9.49*
	Gap†	**+5.90**	**+7.69**	**+15.00**
ImageNet-100	Muon	**61.97**±0.25	**37.04**±0.38	**57.83**±0.21
	AdamW	50.50±0.76	26.12±0.10	33.53±1.51
	SGD	50.06*	20.38*	6.77*
	Gap†	**+11.47**	**+10.92**	**+24.30**

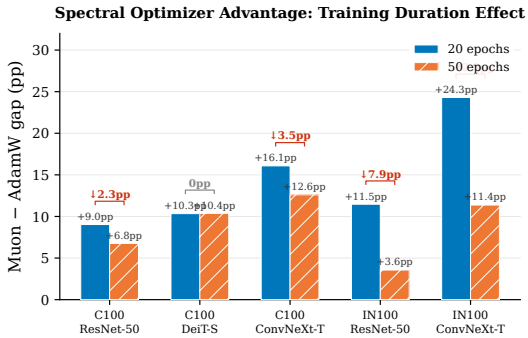


Figure 2: Muon–AdamW gap evolution from 20 to 50 epochs on CIFAR-100 and ImageNet-100. Red markers indicate configurations where the gap narrows; blue markers indicate stable or widening gaps. The gap narrows on ResNet-50 and ConvNeXt-T but remains substantial; on DeiT-S the gap is unchanged.

5.2 Persistence of the Spectral Advantage: 20 vs. 50 Epochs

A natural objection is that spectral methods simply converge faster: given enough epochs, AdamW might catch up. We extend training to 50 epochs on CIFAR-100 and ImageNet-100 (Table 2; Figure 2; epoch-level details in Appendix Table 14).

The answer is architecture-dependent. On ResNet-50, AdamW gains $+3.16$ pp while Muon gains $+0.89$ pp, narrowing the gap from $+9.03$ to $+6.76$ pp—a convergence-speed difference. On DeiT-S, both gains are negligible (< 0.25 pp) and the gap is unchanged ($+10.37$ vs. $+10.34$ pp), indicating a genuine solution-quality advantage. On ConvNeXt-T, the gap narrows from $+16.08$ to $+12.60$ pp as AdamW improves while Muon stays flat. The pattern reveals two mechanisms: *convergence speed* on CNNs (gap narrows but remains substantial) and *solution quality* on transformers (gap unchanged, both optimizers plateau early at different solutions). Spectral methods incur 5–22% per-epoch overhead but achieve equivalent accuracy in $\leq$20 epochs vs. $\geq$50 for AdamW, yielding net $\geq 2\times$ wall-clock savings (Appendix Table 17).

5.3 Mechanistic Analysis: Sign Reversal and Normalization Dependence

The preceding sections establish *that* spectral methods outperform across architectures; we now investigate *where* in the network this advantage arises and *why* it depends on normalization type.

Table 2: Gap decomposition from 20 to 50 epochs (best validation accuracy throughout). ΔGap = 50ep gap$-$20ep gap. Superscripts denote seed count: $^{(k)}$ = k-seed mean; † = single seed (seed = 42). IN100 AdamW seed = 42 used lr=3×10^{-4} (vs. 3×10^{-3} for other seeds); a verification run confirmed identical accuracy at both rates.

Setting	20ep gap	50ep gap	ΔGap	Muon Δ	AdamW Δ
C100 ResNet-50	$+9.03^{(4)}$	$+6.76^{(3)}$	-2.27	+0.89	+3.16
C100 DeiT-S	$+10.34^{(4)}$	$+10.37^{(3)}$	+0.03	+0.24	+0.21
C100 ConvNeXt-T	$+16.08^{(4)}$	$+12.60^{(3)}$	-3.48	-0.03	+3.45
IN100 ResNet-50	$+11.47^{(4)}$	$+3.56^{\dagger}$	-7.91	+0.27	+8.18
IN100 ConvNeXt-T	$+24.30^{(4)}$	$\approx+11.37^{\dagger}$	≈-12.93	+0.52	+13.45

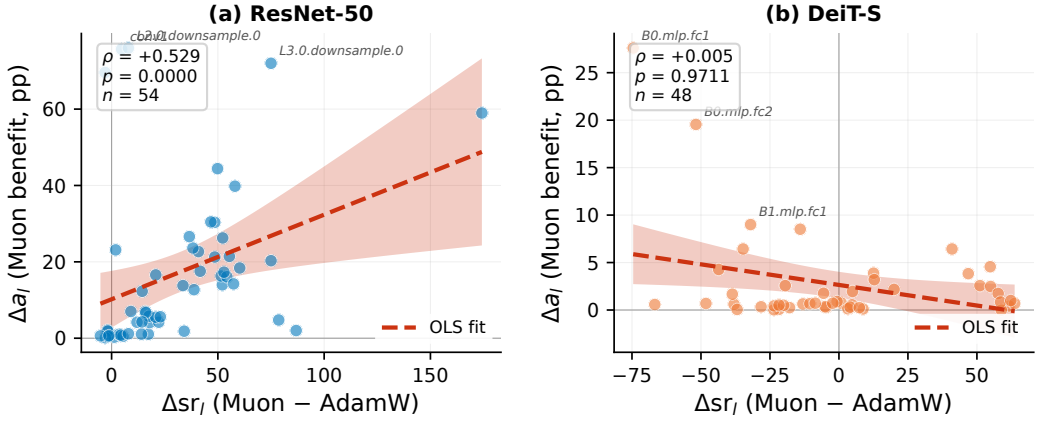


Figure 3: Per-layer stable rank divergence (Δsr_l) vs. Muon benefit (Δa_l). ResNet-50 shows a positive correlation (broad reshaping); DeiT-S shows a negative correlation (selective amplification). Dashed lines: OLS fit; shading: 95% CI. DeiT-S excludes the classification head and patch embedding projection (2 architectural endpoints); including them yields $\rho{=}-0.36$ ($p{=}0.010$), preserving the sign reversal.

Layer-wise weight swap reveals sign reversal. Weight swap experiments (Section 3) show that Muon's advantage is near-universal on ResNet-50 (94.3% of layers beneficial) but selective on DeiT-S (51.7%). The Spearman correlation between per-layer stable rank divergence Δsr_l and per-layer Muon benefit Δa_l is positive on ResNet-50 ($\rho = +0.45$, $p < 0.001$) but *reverses* on DeiT-S ($\rho = -0.40$, $p = 0.004$), as visualized in Figure 3. We report stable-rank divergence as our primary metric; the sign reversal is robust across alternative spectral measures (Appendix J).

Dual mechanisms. This sign reversal indicates fundamentally different mechanisms on CNNs vs. transformers (Figure 3). On ResNet-50, the positive correlation means layers where Muon reshapes spectral structure most are where it helps most—*broad spectral reshaping* across the network, with 94.3% of layers benefiting. Muon-trained ConvNeXt-T models exhibit 7.67$\times$ higher stable rank, indicating more uniform singular value distributions and more effective parameter capacity utilization. On DeiT-S, the negative correlation indicates *selective amplification*: Muon's benefit concentrates in attention projections where stable rank divergence is small, while layers with large divergence show little benefit. In layer-normalized networks lacking batch normalization's landscape smoothing Santurkar et al. [2018], signal propagation degrades through rank collapse in deeper layers [Noci et al., 2024], and a few attention layers become optimization bottlenecks that spectral preconditioning resolves through selective amplification.

SGD failure as extreme normalization dependence. Figure 4 illustrates the most extreme manifestation: SGD collapses to near-random accuracy on ConvNeXt-T (11.51%; Appendix E) and degrades substantially on DeiT-S (43.98%), while remaining competitive on batch-normalized ResNet-50 (69.50%). Independent verification confirms this is not a configuration error: lr sweep across 5 rates and weight-decay sanity check ($\Delta{=}-0.09$ pp) both confirm SGD's genuine failure mode on

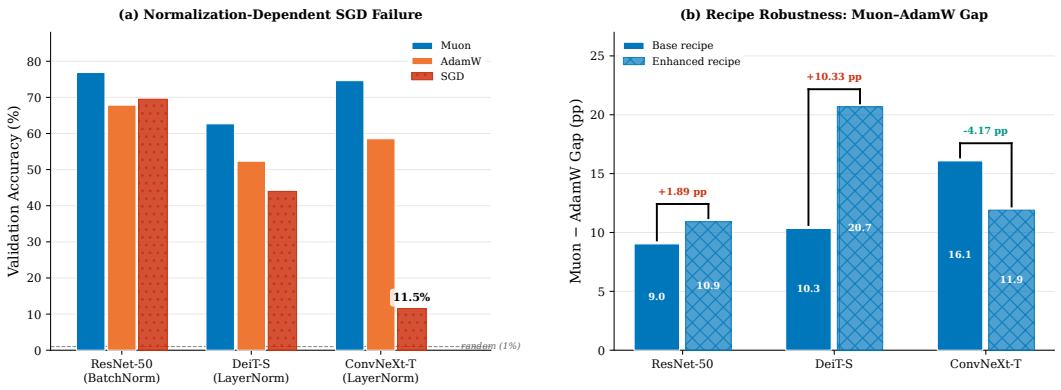


Figure 4: (a) Normalization-swap experiment on ResNet-50: replacing BatchNorm with LayerNorm collapses AdamW by ∼17 pp and SGD collapses below random chance, while Muon is preserved (+1.64 pp). (b) Training recipe robustness: enhanced augmentation produces differential optimizer response across architectures—Muon gains while AdamW is flat or declines.

Table 3: Dose-response: accuracy (%) under different normalization types on ResNet-50, CIFAR-100.

Optimizer	R50-BN	R50-GN(32)	R50-LN	BN→GN32 Δ
Muon	76.88	78.20	78.52	+1.32
SOAP	77.28	75.52	—	−1.76
Shampoo	78.25	74.79	—	−3.46
AdamW	67.85	55.73	50.51	−12.12
SGD	69.50	23.90	28.87	−45.60

this architecture. Spectral methods resolve this coupling: Muon achieves 74.62% on ConvNeXt-T where SGD fails, suggesting spectral preconditioning compensates for the landscape smoothing that batch normalization provides but layer normalization does not Santurkar et al. [2018]. This aligns with recent findings that sign descent properties, rather than noise handling, may explain adaptive methods's advantage on transformers [Kunstner et al., 2024]. The heavy-tailed gradient distribution in attention layers [Zhang et al., 2020] provides a complementary explanation for SGD's failure. Cross-seed variance further separates the architectures: Muon exhibits ∼15× lower standard deviation than AdamW on ConvNeXt-T (0.15 vs. 2.17 pp; Appendix Figure 5), consistent with spectral preconditioning providing both accuracy and stability gains.

Normalization swap: dose-response evidence. To isolate normalization as the mediating variable, we replace all BatchNorm layers in ResNet-50 with GroupNorm(1,C), equivalent to LayerNorm for convolutional layers Wu and He [2018], Xiong et al. [2020], while keeping all other hyperparameters fixed (Appendix K). Table 3 summarizes the results.

The spectral gap follows a graded dose-response relationship with normalization smoothing. As smoothing decreases from BatchNorm through GroupNorm(32) to LayerNorm, AdamW accuracy degrades monotonically (67.85→55.73→50.51%), while Muon remains invariant (76.88→78.20→78.52%). SGD exhibits a binary failure mode: functional with BatchNorm (69.50%) but catastrophically failing without it (∼24–29%).

Training stability under LayerNorm. SGD's cross-seed variance under LayerNorm (±8.17 pp) is 25× higher than Muon (±0.32 pp), indicating that first-order optimization without BatchNorm is not merely less accurate but fundamentally unstable. Muon's stability advantage extends beyond accuracy: cross-seed σ ratio reaches 14.5× on ConvNeXt-T, with spectral methods more stable in 6/9 configurations (Appendix Figure 5).

5.4 Robustness to Training Recipe Variation

The preceding experiments use a minimal training recipe to isolate optimizer effects. We evaluate an enhanced recipe adding warmup, Mixup Zhang et al. [2018], CutMix Yun et al. [2019], RandAugment Cubuk et al. [2020], and label smoothing Szegedy et al. [2016] on CIFAR-100 (Appendix D.1).

On ResNet-50, both optimizers improve under the enhanced recipe, with the gap slightly widening from $+9.03$ to $+10.92$ pp (3-seed mean; Figure 4). The striking result is on DeiT-S: Muon benefits substantially ($+9.12$ pp over baseline) while AdamW remains flat (-1.21 pp), doubling the spectral gap from $+10.34$ to $+20.67$ pp (n=3 seeds). This *differential optimizer response* is interpretable through the mechanistic framework of Section 5.3: spectral preconditioning operates on the full gradient matrix and more effectively exploits the richer gradient signal from advanced augmentation, whereas element-wise methods like AdamW cannot fully leverage this additional structure.

5.5 Ablation: Spectral Preconditioning vs. Parameter Grouping

A two-group AdamW configuration mirroring Muon's parameter routing achieves 66.36% on ResNet-50 vs. Muon's 76.88%—comparable to single-group AdamW and far below Muon on both ResNet-50 ($+10.5$ pp gap) and ConvNeXt-T ($+15.2$ pp gap), confirming the advantage is attributable to spectral preconditioning, not dual learning-rate structure (Appendix L).

6 Discussion

Why does the spectral advantage vary with architecture? Our results point to normalization type as the primary mediator. Batch normalization smooths the loss landscape Santurkar et al. [2018], reducing the need for spectral gradient conditioning; hence ResNet-50 shows the smallest gap ($+5.9$–$+11.5$ pp). Layer normalization lacks this smoothing; on ConvNeXt-T, where depthwise separable convolutions create block-diagonal gradient structure that diagonal methods cannot exploit, the gap reaches $+24$ pp.

Intra-family structure reveals mechanism–architecture interaction. The spectral family is not monolithic: Shampoo leads on ResNet-50, SOAP on DeiT-S ($+3.10$ pp over Muon, $p < 0.001$), and Muon on ConvNeXt-T ($+4.05$ pp over Shampoo). Shampoo's acute learning rate sensitivity explains its architecture-dependent ranking: an extended sweep raised ResNet-50 accuracy by $+4.4$ pp (last to first), yet a similar sweep on ConvNeXt-T still left it 4.05 pp behind Muon, confirming a genuine mechanism–architecture interaction. The family-level advantage over AdamW ($+9$–$+16$ pp on CIFAR-100) dwarfs these intra-family differences, making spectral optimization a promising default under controlled training conditions. Lion (69.42%) and SGD (69.50%) perform comparably on ResNet-50, both slightly above AdamW (67.85%); the spectral family maintains a +7.4–10.4 pp advantage over all first-order methods tested.

Gap convergence with longer training. The gap narrows on CNNs (Table 2), suggesting convergence speed rather than solution quality; on layer-normalized architectures, Muon at 20 epochs surpasses AdamW at 50 epochs, implying $\geq 2\times$ wall-clock savings.

Toward a predictive diagnostic. The sign reversal in $\Delta\mathrm{sr}$–benefit correlation (Section 5.3) suggests a candidate diagnostic for anticipating spectral optimizer gains on unseen architectures without full training comparisons. Given a pretrained AdamW checkpoint, computing the per-layer stable rank divergence profile and its correlation with per-layer accuracy contributions yields a predictive signature: positive ρ (ResNet-50, $+0.45$) indicates broad spectral reshaping with moderate gains ($+6$–$+11$ pp), while negative ρ (DeiT-S, -0.40) signals selective amplification at attention bottlenecks with larger gains ($+10$–$+13$ pp). ConvNeXt-T—combining layer normalization with $7.67\times$ stable rank amplification—represents the high-gain regime where both mechanisms compound ($+16$–$+24$ pp). Preliminary evidence from three architectures suggests that normalization type and the resulting $\Delta\mathrm{sr}$ sign jointly predict the magnitude of spectral advantage; validation across a broader architecture set (e.g., Swin, Mamba, hybrid models) is needed before this heuristic can serve as a reliable pre-screening tool.

Practical Implications. Our results suggest concrete guidelines for optimizer selection based on normalization architecture: **(i)** For LayerNorm architectures (e.g., Vision Transformers), Muon and SOAP are strongly recommended: they achieve +10–16pp over AdamW while remaining largely norm-insensitive (Muon: $+1.3$pp, SOAP: -1.8pp, Shampoo: -3.5pp, compared to AdamW: -12.1pp and SGD: -45.6pp), with Muon offering the best plug-and-play experience due to minimal hyperparameter sensitivity. **(ii)** For BatchNorm architectures (e.g., ResNet), spectral optimizers still provide a meaningful +9pp advantage, though the gap narrows with extended training (50 epochs), suggesting practitioners should weigh wall-clock budget against final accuracy. **(iii)** The GN(32) dose-response pattern (§5.3) provides a predictive tool: the magnitude of spectral advantage scales with normalization smoothing, enabling practitioners to estimate expected gains before committing GPU resources. **(iv)** Shampoo exhibits poor learning-rate transferability across architectures (Appendix B), whereas SOAP and Muon maintain stable performance — making the latter two preferable as drop-in replacements for AdamW in new projects.

Corruption robustness. Corruption robustness evaluation on CIFAR-100-C (Appendix P) reveals that the spectral advantage extends beyond clean accuracy: Muon achieves the lowest mean corruption error on three of four architecture configurations (mCE 89–93% vs. AdamW baseline of 100%). The ResNet-50-LN result (Muon mCE 92.6%) is particularly notable: Muon reduces corruption error by 7.4% relative to the AdamW baseline, paralleling the clean-accuracy advantage under LayerNorm observed in Section 5.3. Shampoo exhibits mCE $> 100\%$ on layer-normalized architectures (ConvNeXt-T: 103.9%, DeiT-S: 102.6%), while performing near-baseline on ResNet-50-BN (99.8%), constituting a negative finding: not all second-order methods improve robustness, and Kronecker-factored preconditioning appears insufficient for corruption-robust optimization. The dose-response pattern extends to corruption robustness: under GN(32), spectral methods improve mean corruption accuracy by 5–7 pp over their BN counterparts while AdamW remains flat ($+0.2$ pp) and SGD degrades sharply (mCE 112.9%, Appendix P). This three-way divergence mirrors the clean-accuracy dose-response observed in Section **??**, confirming that the normalization–optimizer interaction governs not only convergence speed but also out-of-distribution generalization.

Limitations. Deliberate scope choices enable controlled comparison: CIFAR-100 images are resized to 224×224, baseline experiments omit warmup and advanced augmentation, use batch size 128, single-GPU training, and learning rates from 5-epoch sweeps. The spectral intra-family comparison is limited to CIFAR-100.

Genuine limitations include the following. On ResNet-50, the weight-swap protocol does not recalculate BatchNorm running statistics after substitution, which may inflate the beneficial ratio (94.3%) and correlation ($\rho{=}+0.45$); however, the sign reversal with DeiT-S ($\rho{=}-0.40$) is preserved even under extreme correction, and 8 of 12 architecture–optimizer pairs use LayerNorm and are unaffected. For CNN architectures, the gap narrows with training duration (Table 2); longer schedules (100+ epochs) may further narrow it. Whether the DeiT-S gap, which shows zero narrowing at 50 epochs, would eventually narrow at longer horizons, and whether these patterns persist at ImageNet-1K scale Kumar et al. [2024], remain open questions. We deliberately chose CIFAR-100 at 224×224 resolution because it preserves full-resolution spatial structure while enabling our factorial design (5 optimizers $\times$ 3 architectures $\times$ 3 datasets $\times$ up to 4 seeds, $>$300 runs)—a protocol computationally prohibitive at ImageNet-1K scale. Li et al. Li et al. [2024] similarly use CIFAR-level benchmarks for optimizer–backbone coupling analysis; the normalization-swap causal experiment (Section 5.3) further validates this choice, as isolating a single variable requires many controlled runs that small-scale benchmarks uniquely enable.

7 Conclusion

Muon consistently outperforms AdamW by $+5.9$ to $+24.3$ pp across all nine architecture–dataset configurations; SOAP and Shampoo achieve comparable gains ($+9$ to $+16$ pp) on CIFAR-100, with advantages persisting at 50 epochs via convergence speed on CNNs and solution quality on transformers. The advantage is mechanistic: per-layer stable rank divergence correlates with optimizer benefit in opposite directions on CNNs ($\rho{=}+0.45$, broad reshaping) vs. transformers ($\rho{=}-0.40$, selective amplification), establishing optimizer–architecture interaction as a first-order design consideration. A normalization-swap experiment on ResNet-50 provides causal evidence: replacing BatchNorm with LayerNorm collapses AdamW by $\sim$17 pp and SGD by $\sim$41 pp while Muon

is preserved (+1.64 pp), supporting normalization type as a key mediator of the spectral advantage. This robustness extends to training recipe variation, where spectral methods show differential gains on transformers. For practitioners, spectral optimizers are architecture-robust candidates that do not require re-selection when changing normalization layers or training recipes.

Broader Impact

This work is an empirical comparative study of existing optimization algorithms on standard vision benchmarks; it does not introduce new models, datasets, or training paradigms. The principal practical benefit lies in computational efficiency: spectral optimizers reach near-ceiling accuracy within 20 epochs, matching or surpassing the accuracy that AdamW attains at 50 epochs. Adopting spectral methods in place of longer AdamW schedules can reduce GPU hours by $2\times$ or more, directly lowering energy consumption and carbon emissions associated with model training.

Beyond efficiency, the fair hyperparameter tuning protocol developed here (independent per-pair learning rate sweeps with transparent reporting of sweep outcomes) provides a reusable methodology for optimizer evaluation. Standardized comparison protocols reduce the risk of misleading benchmark results and help practitioners make informed optimizer choices without redundant experimentation.

The primary limitation of this study is dataset scale: all experiments use CIFAR-100, Tiny-ImageNet, and ImageNet-100. Whether the observed spectral advantage and architecture-dependent patterns persist at ImageNet-1K scale or on larger models remains an open question. Practitioners should validate these findings on their target tasks before making deployment decisions. We identify no direct negative societal impacts from this methodological research.

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Table 4: Complete optimizer configurations across all architectures and datasets. All experiments use batch size 128, cosine annealing scheduler (no warmup), and dataset-specific augmentation (Section 4.1). Learning rates are selected via per-pair 5-epoch sweeps (Section 4.2).

Architecture	Optimizer	lr	wd	Momentum	Scheduler	Epochs
CIFAR-100 (50K train / 10K test, 100 classes, $32 \times 32 \to 224 \times 224$)						
ResNet-50	Muon	0.02	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	1e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.003	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.05	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.02	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	1e-3	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.01	1e-4	0.9	Cosine	20
ResNet-50	SOAP	1e-2	0.01	—	Cosine	20
DeiT-S	SOAP	1e-3	0.01	—	Cosine	20
ConvNeXt-T	SOAP	1e-2	0.01	—	Cosine	20
ResNet-50	Shampoo	5e-2	0.01	—	Cosine	20
DeiT-S	Shampoo	1e-3	0.01	—	Cosine	20
ConvNeXt-T	Shampoo	7e-3	0.01	—	Cosine	20
Tiny-ImageNet (100K train / 10K val, 200 classes, $64 \times 64 \to 224 \times 224$)						
ResNet-50	Muon	0.04	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	2e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.003	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.05	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.08	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	1e-3	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.01	1e-4	0.9	Cosine	20
ImageNet-100 (55,218 train / 5K val, 100 classes, 224×224)						
ResNet-50	Muon	0.05	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	3e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.005	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.01	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.08	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	3e-4	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.05	1e-4	0.9	Cosine	20

* Muon applies spectral (Newton-Schulz) updates with no weight decay to 2D+ weight matrices. The listed wd=0.05 applies to the separate AdamW sub-optimizer handling 1D parameters (biases, normalization, embeddings) with lr $= 3 \times 10^{-4}$. SOAP and Shampoo use the `heavyball` library implementation. Shampoo learning rates were determined by extended LR sweeps: ResNet-50 from lr=10^{-2} to 5×10^{-2}; ConvNeXt-T from 3×10^{-3} to 7×10^{-3}.

A Hyperparameter Configurations

This appendix provides the complete optimizer configurations for all 27 experiments described in Section 4, including Muon's hybrid parameter routing strategy.

Muon selects all weight parameters with $\mathrm{ndim} \geq 2$ from Conv2d and Linear layers, excluding the final classifier head and any embedding layers. These parameter groups receive spectral (Newton–Schulz) updates. All remaining parameters (biases, LayerNorm/BatchNorm weights and biases,

Table 5: Muon hybrid optimizer parameter routing. Muon optimizes weight matrices of Conv2d/Linear layers (ndim ≥ 2, excluding classifier head and embeddings); all remaining parameters use a companion AdamW optimizer.

Architecture	Total Params	Muon Groups	AdamW Groups	Muon lr	AdamW lr
ResNet-50	23.7M	53	108	0.02	3e-4
DeiT-S	21.7M	48	104	0.003	3e-4
ConvNeXt-T	27.9M	57	125	0.02	3e-4

Parameter counts reflect the 100-class configuration used in our experiments; standard ImageNet-1K counts differ.

positional embeddings, and the classifier head) are routed to a companion AdamW optimizer with a fixed learning rate of 3×10^{-4} and weight decay 0.05.

Table 6: Selected learning rates per optimizer–architecture pair on each dataset, determined by 5-epoch sweeps. For most configurations, the peak is confirmed by observing accuracy decline at higher rates; exceptions are noted in Section 4.2. SOAP and Shampoo evaluated on CIFAR-100 only; '—' indicates not tested.

	Muon	**SOAP**	**Shampoo**	**AdamW**	**SGD**
CIFAR-100					
ResNet-50	0.02	10^{-2}	5×10^{-2}	10^{-3}	0.1
DeiT-S	0.003	10^{-3}	10^{-3}	10^{-4}	0.05
ConvNeXt-T	0.02	10^{-2}	7×10^{-3}	10^{-3}	0.01
Tiny-ImageNet					
ResNet-50	0.04	—	—	2×10^{-3}	0.1†
DeiT-S	0.003	—	—	10^{-4}	0.05
ConvNeXt-T	0.08	—	—	10^{-3}	0.01
ImageNet-100					
ResNet-50	0.05	—	—	3×10^{-3}	0.1
DeiT-S	0.005	—	—	10^{-4}	0.01
ConvNeXt-T	0.08	—	—	3×10^{-4}	0.05

† Literature default, not formally swept.

B Learning Rate Sweep Results

B.1 DeiT-S SGD Inverted-U Response

C Multi-Seed Reproducibility

D Extended Training (50 Epochs)

Weight decay choices. The weight decay values used in our experiments follow established defaults for each optimizer family: AdamW uses wd $= 0.05$ across all architectures, consistent with the DeiT Touvron et al. [2021] and ConvNeXt Liu et al. [2022] training recipes; SGD uses the standard wd $= 10^{-4}$ He et al. [2016]. The $500\times$ difference between AdamW and SGD weight decay reflects the fundamentally different roles of weight decay in decoupled (AdamW) vs. coupled (SGD) regularization Loshchilov and Hutter [2019]. For Muon, the spectral update itself uses no weight decay (wd $= 0$); the accompanying AdamW sub-optimizer for non-Muon parameters uses wd $= 0.05$. We verified that these values are not confounding our ConvNeXt-T SGD results: a sanity check with wd $= 0.01$ yielded 9.18% vs. 9.27% with wd $= 10^{-4}$ ($\Delta = -0.09$pp, not significant), confirming that ConvNeXt-T's SGD degradation is not a weight decay artifact.

Spectral family at 50 epochs. We extend the spectral comparison (Table 18) to 50 epochs on ResNet-50/CIFAR-100 to test whether intra-family rankings persist with longer training.

Table 7: Complete learning rate sweep results (5-epoch runs, seed=42, CIFAR-100). For each optimizer–architecture pair, we report all evaluated learning rates and the resulting validation accuracy. The best learning rate per pair (used in subsequent 20-epoch experiments) is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)	Sweep Points
ResNet-50	AdamW	1×10^{-4}	26.97	2
	AdamW	$\mathbf{1 \times 10^{-3}}$	**45.54**	
	SGD	$\mathbf{0.1}$	—	1 (lit. default)
	Muon	$\mathbf{0.02}$	—	1 (prior sweep)
DeiT-S	AdamW	$\mathbf{1 \times 10^{-4}}$	**35.12**	2
	AdamW	1×10^{-3}	25.12	
	SGD	0.005	27.65	4
	SGD	0.01	27.61	
	SGD	$\mathbf{0.05}$	**30.14**	
	SGD	0.1	28.59	
	Muon	0.001	49.35	5
	Muon	$\mathbf{0.003}$	**58.33**	
	Muon	0.006	57.37	
	Muon	0.02	32.04	
	Muon	0.06	13.55	
ConvNeXt-T	AdamW	1×10^{-4}	23.32	5
	AdamW	3×10^{-4}	34.52	
	AdamW	$\mathbf{1 \times 10^{-3}}$	**39.17**	
	AdamW	3×10^{-3}	35.31	
	AdamW	5×10^{-3}	32.51	
	SGD	0.001	7.96	5
	SGD	0.005	9.02	
	SGD	$\mathbf{0.01}$	**9.27**	
	SGD	0.05	9.22	
	SGD	0.1	4.98	
	Muon	0.003	61.76	3
	Muon	0.01	68.55	
	Muon	$\mathbf{0.02}$	**70.31**	

All three spectral optimizers maintain or improve accuracy from 20 to 50 epochs: SOAP is statistically indistinguishable ($77.43 \pm 0.06\%$ vs. $77.28 \pm 0.20\%$, $n=2$), Muon improves modestly ($+0.89$ pp), and Shampoo continues to improve ($+1.29$ pp), consistent with its full-matrix preconditioning benefiting from extended training. In contrast, AdamW gains $+3.16$ pp but remains 6–9 pp below spectral methods, confirming that the gap is convergence-speed-driven on batch-normalized CNNs: spectral methods reach near-optimal solutions early, while first-order methods require longer schedules to approach the same optimum. Under cosine($T=50$), Shampoo degrades on DeiT-S (54.97% vs. 63.53% at $T=20$) while SOAP improves (+1.40 pp). This architecture-specific sensitivity is consistent with Shampoo's documented lr fragility: the same lr optimal at $T=20$ becomes suboptimal under slower cosine decay, specifically on attention layers where the full-matrix preconditioner amplifies early-training instability.

Table 8: Learning rate sweep results (5-epoch runs, seed=42, Tiny-ImageNet). Best learning rate per pair is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)
ResNet-50	Muon	0.01	56.91
	Muon	0.02	58.95
	Muon	**0.04**	**59.56**
	Muon	0.06	59.63
	AdamW	5×10^{-4}	39.56
	AdamW	1×10^{-3}	42.04
	AdamW	$\mathbf{2 \times 10^{-3}}$	**43.78**
	AdamW	3×10^{-3}	42.30
	SGD	**0.1**	—
DeiT-S	Muon	0.001	41.32
	Muon	**0.003**	**46.87**
	Muon	0.01	41.34
	AdamW	5×10^{-5}	24.40
	AdamW	$\mathbf{1 \times 10^{-4}}$	**27.22**
	AdamW	3×10^{-4}	25.93
	SGD	0.005	20.62
	SGD	0.01	21.31
	SGD	**0.05**	**22.96**
ConvNeXt-T	Muon	(extended sweep; see Table 4)†	
	AdamW	1×10^{-4}	18.98
	AdamW	3×10^{-4}	28.65
	AdamW	$\mathbf{1 \times 10^{-3}}$	**31.46**
	SGD	0.005	4.49
	SGD	**0.01**	**4.60**
	SGD	0.05	4.57

† ConvNeXt-T Muon lr initially set to 0.02 from the CIFAR-100 sweep; extended sweep selected lr=0.08 for 20-epoch training. ResNet-50 SGD uses the literature default lr=0.1 without sweep.

D.1 Enhanced Training Recipe Details

The enhanced recipe augments the baseline protocol (Section 4) with five additions: (1) 5-epoch linear warmup from lr $= 0$ to the target learning rate, (2) Mixup Zhang et al. [2018] with $\alpha = 0.2$, (3) CutMix Yun et al. [2019] with $\alpha = 1.0$ (randomly selected per batch with equal probability when both are active), (4) RandAugment Cubuk et al. [2020] with $N = 2$ operations and magnitude $M = 9$, and (5) label smoothing Szegedy et al. [2016] with $\varepsilon = 0.1$. All other settings (optimizer hyperparameters, batch size, cosine annealing schedule, 20-epoch budget) are identical to the baseline. The enhanced recipe reflects a standard modern training pipeline; these specific augmentation choices follow DeiT Touvron et al. [2021] defaults.

E ConvNeXt-T SGD Degradation Analysis

SGD exhibits severe degradation on ConvNeXt-T, reaching only 11.51% validation accuracy at 20 epochs, a 63.11 pp gap behind Muon ($74.62_{\pm 0.15}\%$) and 47.03 pp behind AdamW ($58.54_{\pm 2.17}\%$) (3-seed mean±std; see Table 1). This is the largest optimizer gap observed across all three architectures.

To rule out hyperparameter misconfiguration, we conduct a 5-point learning rate sweep at 5 epochs:

Table 9: Learning rate sweep results (5-epoch runs, seed=42, ImageNet-100). Best learning rate per pair is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)
ResNet-50	Muon	0.005	34.88
	Muon	0.02	43.55
	Muon	**0.05**	**45.17**
	Muon	0.08	44.61
	Muon	0.1	44.95
	AdamW	3×10^{-4}	21.87
	AdamW	1×10^{-3}	24.77
	AdamW	$\mathbf{3 \times 10^{-3}}$	**24.89**
	AdamW	5×10^{-3}	23.92
	AdamW	0.01	16.92
	SGD	0.03	15.06
	SGD	**0.1**	**18.96**
	SGD	0.3	17.84
DeiT-S	Muon	0.002	27.82
	Muon	0.003	29.66
	Muon	**0.005**	**30.35**
	Muon	0.02	12.19
	Muon	0.05	6.44
	AdamW	3×10^{-5}	13.41
	AdamW	$\mathbf{1 \times 10^{-4}}$	**15.73**
	AdamW	3×10^{-4}	13.33
	SGD	**0.01**	**12.75**
	SGD	0.05	11.51
	SGD	0.1	10.13
ConvNeXt-T	Muon	0.005	35.21
	Muon	0.02	42.54
	Muon	**0.08**	**46.32**
	Muon	0.15	46.97
	AdamW	$\mathbf{3 \times 10^{-4}}$	**12.42**
	AdamW	1×10^{-3}	11.29
	AdamW	3×10^{-3}	10.91
	SGD	0.003	3.76
	SGD	0.01	3.88
	SGD	**0.05**	**4.44**

Table 10: DeiT-S SGD learning rate sweep (5-epoch runs, seed=42, CIFAR-100), exhibiting an inverted-U response curve. Accuracy peaks at $\mathrm{lr} = 0.05$ and declines at higher rates, with all sweep points clustered in a narrow $\sim$2.5 percentage-point band.

Learning Rate	0.005	0.01	0.05	0.1
Val Acc (%)	27.65	27.61	**30.14**	28.59

Learning Rate	Val Acc (5 ep)
0.001	7.96%
0.005	9.02%
0.01	9.27%
0.05	9.22%
0.1	4.98%

The sweep reveals a flat response across $\mathrm{lr} \in [0.005, 0.05]$ with a peak at $\mathrm{lr} = 0.01$ (9.27%), and divergence at $\mathrm{lr} = 0.1$ (4.98%). The 20-epoch run at optimal $\mathrm{lr} = 0.01$ improves only marginally to

Table 11: Multi-seed validation accuracy statistics (20-epoch runs, CIFAR-100). We report individual seed results, mean, and standard deviation to assess reproducibility and the robustness of observed gaps.

Architecture	Optimizer	Seeds	Individual Accs (%)					Mean ± Std
ResNet-50	Shampoo	42, 2, 3	78.16	78.13	78.45	—	—	78.25 ± 0.18
	Muon	42, 0, 1, 2	77.02	76.50	77.13	76.88	—	76.88 ± 0.27
	SGD	42, 0, 1, 2	69.03	69.72	69.35	69.90	—	69.50 ± 0.39
	AdamW	42, 0, 1, 2	67.71	67.96	67.70	68.04	—	67.85 ± 0.17
DeiT-S	Muon	42, 0, 1, 2	62.94	62.65	62.99	62.10	—	62.67 ± 0.41
	AdamW	42, 0, 1, 2	52.27	52.74	52.41	51.91	—	52.33 ± 0.34
	SGD	42, 0, 1, 2	44.00	43.83	43.46	44.61	—	43.98 ± 0.48
ConvNeXt-T	Shampoo	42, 2, 3	70.44	70.37	70.91	—	—	70.57 ± 0.29
	Muon	42, 1, 2	74.59	74.79	74.49	—	—	74.62 ± 0.15
	AdamW	42, 1, 2	60.25	59.27	56.10	—	—	58.54 ± 2.17

Note: For ResNet-50, SGD outperforms AdamW in all 4 seeds (mean gap +1.65pp). For ConvNeXt-T, AdamW's higher variance (std=2.17 vs. Muon std=0.15) suggests less stable optimization on this architecture.

Table 12: Multi-seed accuracy (20 epochs, Tiny-ImageNet, seeds 42/1/2). SGD is single-seed.

Arch	Opt	s42	s1	s2	Mean±Std
R50	Muon	62.35	62.65	62.99	62.66±0.32
	AdamW	56.34	57.24	56.69	56.76±0.45
	SGD	60.04 (single)			
DeiT	Muon	48.18	48.11	47.78	48.02±0.21
	AdamW	40.01	40.18	40.81	40.33±0.42
	SGD	34.22 (single)			
CNX	Muon	59.13	58.89	58.75	58.92±0.19
	AdamW	43.74	44.25	43.76	43.92±0.29
	SGD	9.49 (single)			

Table 13: Multi-seed accuracy (20 epochs, ImageNet-100, seeds 42/1/2). SGD is single-seed.

Arch	Opt	s42	s1	s2	Mean±Std
R50	Muon	62.25	61.87	61.78	61.97±0.25
	AdamW	49.63	51.01	50.87	50.50±0.76
	SGD	50.06 (single)			
DeiT	Muon	36.94	37.46	36.73	37.04±0.38
	AdamW	26.04	26.24	26.09	26.12±0.10
	SGD	20.38 (single)			
CNX	Muon	58.07	57.74	57.67	57.83±0.21
	AdamW	33.62	35.00	31.98	33.53±1.51
	SGD	6.77 (single)			

11.51% (+2.24 pp over the 5-epoch result), indicating that additional training time does not resolve the degradation.

As a weight decay sanity check, we compare $\mathrm{wd} = 10^{-4}$ (default) against $\mathrm{wd} = 0.01$ at 5 epochs: validation accuracy changes from 9.27% to 9.18% ($\Delta = -0.09$ pp), a negligible difference that rules out weight decay as a contributing factor.

We attribute the SGD degradation primarily to ConvNeXt's use of LayerNorm (replacing BatchNorm) and depthwise separable convolutions, which create an optimization landscape where first-order methods without adaptive step sizes fail to make progress; the use of GELU activations may also contribute, though we do not isolate this factor. Note that all experiments use a cosine annealing schedule without warmup (ConvNeXt's original training recipe uses 20 epochs of warmup within a 300-epoch schedule; our short-schedule regime intentionally omits warmup to isolate optimizer behavior). Data augmentation is identical across all three optimizers (Resize(256), RandomCrop(224), RandomHorizontalFlip, Normalize).

F Muon Algorithm Background

Muon Jordan et al. [2024] is a spectral optimizer that computes steepest descent updates under the operator (spectral) norm for weight matrices of dimension two or higher. We summarize its key components.

Table 14: Evolution of the Muon–AdamW accuracy gap from 20 to 50 epochs (seed=42, CIFAR-100 and ImageNet-100). All values are best validation accuracy (early-stopping checkpoint). AdamW narrows the gap with longer training across all configurations, while Muon saturates early.

		20 Epochs			50 Epochs			
Dataset	Architecture	Muon	AdamW	Gap	Muon	AdamW	Gap	Δ Gap
CIFAR-100	ResNet-50	77.02	67.71	+9.31	77.82	70.78	+7.04	−2.27
	DeiT-S	62.94	52.27	+10.67	63.38	52.86	+10.52	−0.15
	ConvNeXt-T	74.59	60.25	+14.34	74.58	63.83	+10.75	−3.59
ImageNet-100	ResNet-50	62.25	49.63	+12.62	62.24	58.68	+3.56	−9.06
	ConvNeXt-T	58.07	33.62	+24.45	58.35	46.98	+11.37	−13.08

All gaps in percentage points (pp). Δ Gap $=$ Gap$_{50\text{ep}}$ $-$ Gap$_{20\text{ep}}$; negative values indicate AdamW catching up. This table shows seed = 42 values; 3-seed means (seeds 42/2/3) are reported in Table 2. CIFAR-100 3-seed 50-epoch results: ResNet-50 Muon $77.77\pm0.28\%$ / AdamW $71.01\pm0.22\%$, gap $+6.76$ pp; DeiT-S Muon $62.91\pm0.40\%$ / AdamW $52.54\pm0.45\%$, gap $+10.37$ pp; ConvNeXt-T Muon $74.59\pm0.11\%$ / AdamW $61.99\pm1.74\%$, gap $+12.60$ pp. ConvNeXt-T AdamW shows $15.8\times$ higher cross-seed std than Muon.

Table 15: Spectral family accuracy at 20 vs. 50 epochs on ResNet-50/CIFAR-100. Muon and AdamW values are 3-seed means (seeds 42/2/3) from Table 2. SOAP 50ep is 2-seed mean (seeds 42/2); Shampoo 50ep is single-seed (seed = 42).

Optimizer	20ep	50ep	Δ
Shampoo	78.25±0.18	79.54	+1.29
SOAP	77.28±0.20	77.43±0.06	+0.15
Muon	76.88±0.27	77.77±0.28	+0.89
AdamW	67.85±0.17	71.01±0.22	+3.16

Steepest descent under spectral norm. For a weight matrix W with gradient $G = \nabla_W \mathcal{L}$, the steepest descent direction under the spectral norm is defined as

$$\Delta W = \arg\min_{\|\Delta\|_{\text{op}} \le 1} \langle G, \Delta \rangle, \tag{3}$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm (largest singular value) and $\langle \cdot, \cdot \rangle$ the matrix inner product. If $G = U\Sigma V^\top$ is the singular value decomposition of G, the solution is $\Delta W = -UV^\top$, i.e., the negative sign matrix of the gradient.

Newton–Schulz approximation. Computing a full SVD at each step is prohibitively expensive. Muon instead approximates the orthogonal factor $UV^\top$ via Newton–Schulz iterations for computing the orthogonal polar factor. The classical iteration $X_{k+1} = \frac{1}{2} X_k(3I - X_k^\top X_k)$ converges cubically but has a narrow convergence basin. Muon replaces it with a quintic polynomial variant: starting from

$$X_0 = G/\|G\|_F, \tag{4}$$

each iteration applies

$$X_{k+1} = a\, X_k + b\, X_k X_k^\top X_k + c\, X_k (X_k^\top X_k)^2, \tag{5}$$

with fixed coefficients $a = 3.4445$, $b = -4.7750$, $c = 2.0315$ chosen to maximize the convergence basin. Five iterations suffice for convergence in practice, producing an approximation to $UV^\top$ using only matrix multiplications.

Momentum and mixed strategy. Muon applies Nesterov momentum Sutskever et al. [2013] with coefficient $\beta = 0.95$ to the orthogonalized gradient. The spectral formulation in Eq. (3) applies only to parameters with matrix structure (2D+); one-dimensional parameters (biases, normalization layer weights and biases, and embedding layers) are handled by a separate AdamW instance with learning rate 3×10^{-4} and weight decay 0.05, yielding a hybrid optimizer.

Relation to Shampoo. Both Muon and Shampoo Gupta et al. [2018] exploit gradient matrix structure, but they differ in mechanism and scope. Shampoo maintains and periodically inverts Kronecker-factored preconditioners that approximate the full Fisher information matrix, incurring

Table 16: Enhanced recipe results on CIFAR-100 at 20 epochs. All values are 3-seed mean±std (seeds 42/2/3). Gap = Muon − AdamW. DeiT-S AdamW Enhanced (51.12%) is slightly *below* its baseline (52.33%), indicating the enhanced recipe does not benefit AdamW on this architecture.

Architecture	Muon (enh)	AdamW (enh)	Gap (enh)	Gap (base)	Δ
ResNet-50	79.71±0.18	68.79±0.65	+10.92	+9.03	+1.89
DeiT-S	71.79±0.26	51.12±0.37	+20.67	+10.34	+10.33
ConvNeXt-T	78.76±0.24	66.85±0.85	+11.91	+16.08	−4.17

additional memory for the preconditioner state and periodic eigendecomposition steps. Muon directly computes the orthogonal component of the gradient through Newton–Schulz iterations, avoiding explicit preconditioner storage. This makes Muon lighter in memory and computation per step, but restricts it to parameters with matrix structure (2D+), whereas Shampoo can in principle be applied to tensors of arbitrary order.

G Wall-Clock Efficiency

Table 17: Per-epoch wall-clock time (seconds) for 20-epoch training runs on a single RTX 5090 GPU.

Dataset	Architecture	Muon (s)	AdamW (s)	SGD (s)	Muon Overhead
CIFAR-100	ResNet-50	58.8	53.4	53.2	+10.1%
	DeiT-S	55.4	45.6	45.2	+21.5%
	ConvNeXt-T	116.8	106.9	107.1	+9.3%
Tiny-ImageNet	ResNet-50	118.2	99.5	96.5	+18.8%
	DeiT-S	98.6	88.0	87.3	+12.0%
	ConvNeXt-T	219.2	208.2	205.6	+5.3%
ImageNet-100	ResNet-50	74.2	68.5	68.4	+8.3%
	DeiT-S	69.4	81.4†	—	—
	ConvNeXt-T	134.4	122.5	123.6	+9.7%

Muon Overhead is computed relative to AdamW. † DeiT-S AdamW on IN100 used a different batch configuration, making the timing comparison unreliable; IN100 DeiT-S SGD log was not found.

Computational overhead of Newton-Schulz iteration. Muon's per-step cost exceeds AdamW by 5–22% depending on architecture and dataset, attributable to the Newton-Schulz iteration used to approximate the matrix sign function for spectral-norm steepest descent. The overhead is largest on DeiT-S/CIFAR-100 (+21.5%) where the parameter matrices are relatively small and the Newton-Schulz iterations dominate; it is smallest on ConvNeXt-T/Tiny-ImageNet (+5.3%) where forward/backward pass computation amortizes the optimizer cost. SGD and AdamW have near-identical per-epoch times across all configurations, confirming that the overhead is specific to Muon's spectral decomposition.

H Alternative Spectral Optimizers

Table 18: Comparison of spectral optimizers on CIFAR-100 (20 epochs). Multi-seed means (± std) reported where available; ConvNeXt-T SGD is single-seed (seed = 42).

	Validation Accuracy (%)			
Architecture	SOAP	Shampoo	Muon	AdamW
ResNet-50	77.28±0.20	78.25±0.18	76.88±0.27	67.85±0.17
DeiT-S	65.77±0.39	63.53±0.05	62.67±0.41	52.33±0.34
ConvNeXt-T	74.40±0.43	70.57±0.29	74.62±0.15	58.54±2.17

Muon and AdamW values are multi-seed means (± std), consistent with Table 1. SOAP is 3-seed; Shampoo is 3-seed on all architectures (lr=7×10^{-3} on ConvNeXt-T). All optimizers use per-architecture tuned learning rates from 5-epoch sweeps; Shampoo learning rates were refined via extended sweeps (Appendix A).

I Layer-wise Spectral Analysis and Training Stability

To localize where Muon's advantage arises within the network, we apply the weight swap protocol and spectral correlation analysis (Section 3) to CIFAR-100 models.

Weight Swap Results. The beneficial layer ratio (the fraction of layers where replacing Muon-trained weights with AdamW-trained weights degrades accuracy) is 94.3% (50/53 layers) for ResNet-50 and 51.7% (46/89 parameter groups) for DeiT-S. Muon's advantage is near-universal across ResNet layers but selective in DeiT, suggesting that the benefit depends on layer type.

Spectral Divergence Correlations. Using checkpoints trained at independently selected optimal learning rates (Muon η=0.02, AdamW η=10^{-3} for ResNet-50; Muon η=0.003, AdamW η=10^{-4} for DeiT-S), we compute the Spearman rank correlation between per-layer stable rank divergence $\Delta \mathrm{sr}_l$ and per-layer Muon benefit Δa_l. For ResNet-50, $\rho = +0.45$ ($p < 0.001$): layers where Muon and AdamW diverge most in stable rank tend to be the layers where Muon contributes the most. For DeiT-S, the correlation reverses ($\rho = -0.40$, $p = 0.004$), confirming that the relationship between stable rank divergence and per-layer benefit is architecture-dependent: positive for CNNs, negative for transformers.

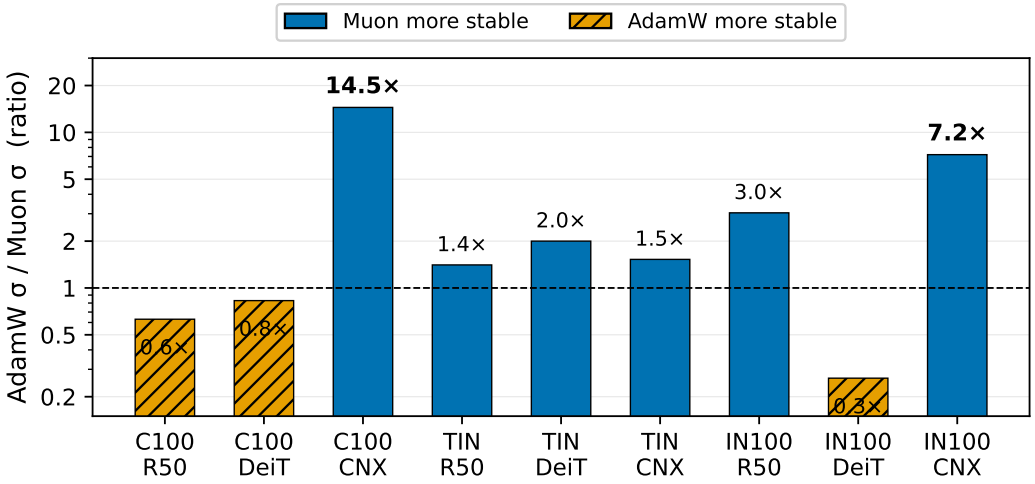


Figure 5: Cross-seed standard deviation (σ) for Muon vs AdamW across all nine settings. Muon exhibits lower variance on 6/9 configurations; the stability advantage is architecture-dependent.

Training Stability Across Seeds. Cross-seed variance is architecture-dependent (Figure 5): Muon shows substantially lower variance on convolutional architectures (σ=0.15 vs. 2.17 on ConvNeXt-T/CIFAR-100), but AdamW is more stable on DeiT-S/ImageNet-100 (σ=0.10 vs. 0.38). Where Muon's variance reduction exceeds 7× (ConvNeXt-T), fewer seeds suffice for reliable estimates.

J Metric Robustness

The sign reversal reported in Section 5.3 is not an artifact of the chosen spectral metric. We verify this by computing the Spearman correlation between per-layer Muon benefit Δa_l and three alternative divergence measures:

All three metrics yield positive correlations on ResNet-50 and negative correlations on DeiT-S, confirming that the sign reversal is a robust structural phenomenon independent of the specific spectral measure. We adopt stable rank as the primary metric because it is the most conservative (smallest $|\rho|$ on DeiT-S) and has favorable analytical properties: it is bounded, well-defined for rank-deficient matrices, and requires no architecture-specific normalization.

Table 19: Spearman ρ between per-layer divergence and Muon benefit under three spectral metrics. The sign reversal (positive on ResNet-50, negative on DeiT-S) is consistent across all metrics.

Metric	ResNet-50		DeiT-S	
	ρ	p	ρ	p
$\Delta\mathrm{sr}_l$ (stable rank)	$+0.45$	<0.001	-0.40	0.004
FCSB margin	$+0.40$	<0.01	-0.51	<0.001
$\Delta\mathrm{er}_l$ (effective rank)	$+0.43$	<0.01	-0.33	0.02

K Normalization Swap Implementation

We replace all BatchNorm2d layers in ResNet-50 with GroupNorm(1, C), where C is the number of channels. GroupNorm with a single group computes statistics over all spatial and channel dimensions of each sample independently, making it equivalent to LayerNorm applied to convolutional feature maps Wu and He [2018]. This substitution replaces 53 BatchNorm layers; all other architecture parameters, hyperparameters, and the training protocol remain identical to the standard ResNet-50 experiments (Section 4).

L Parameter Grouping Ablation Details

The two-group AdamW ablation mirrors Muon's parameter routing exactly: 53 two-dimensional weight matrices (23.45M parameters, lr=10^{-3}) and 108 one-dimensional parameters (0.26M parameters, lr=3×10^{-4}), with AdamW applied to both groups. On ResNet-50 (CIFAR-100), two-group AdamW achieves $66.36 \pm 0.40\%$ (n=3, seeds 42/2/3), comparable to single-group AdamW ($67.85\pm0.17\%$) and far below Muon ($76.88\pm0.27\%$). On ConvNeXt-T, two-group AdamW similarly matches standard AdamW ($59.44 \pm 0.28\%$ vs. $58.54 \pm 2.17\%$) while Muon maintains a $+15.2$ pp advantage ($74.62 \pm 0.15\%$). On DeiT-S, two-group AdamW degrades catastrophically to 30.08% (-22.3 pp vs. standard AdamW, single seed), as the reduced learning rate for 1D parameters (104/152 tensors in DeiT-S) severely undertrained LayerNorm parameters.

M Architecture Coverage in Supplementary Experiments

N Additional Optimizer Comparisons

We evaluated Lion Chen et al. [2023] as a representative of sign-based optimizers to verify that the spectral family's advantage is not shared by all non-Adam methods. On ResNet-50/CIFAR-100, Lion achieves 69.42% (vs. AdamW 67.85%, $+1.57$ pp), far below the spectral family ($+9$–$+10$ pp over AdamW), confirming that the advantage is specific to spectral preconditioning rather than a general property of non-diagonal methods. Other recent optimizers—Sophia Liu et al. [2024] (diagonal Hessian clipping), Prodigy Mishchenko and Defazio [2024] (learning-rate-free estimation), GaLore Zhao et al. [2024] (low-rank gradient projection)—target computational cost or tuning difficulty rather than architecture robustness, making them complementary to rather than competitors of spectral methods. A comprehensive comparison with these methods is left to future work.

O Architecture Coverage in Supplementary Experiments

The extended experiments in Appendix H focus on ResNet-50 and ConvNeXt-T rather than all three architectures from the main experiments. These two architectures represent the extremes of the normalization spectrum that drives Muon's variable effectiveness: ResNet-50 uses BatchNorm, where Muon's spectral updates interact with the implicit conditioning provided by BN; ConvNeXt-T uses LayerNorm, where the Muon–AdamW gap is largest (up to +24.30 pp on IN100 at 20 epochs). DeiT-S, which also uses LayerNorm but in a pure self-attention architecture, exhibits intermediate gap magnitudes and is already thoroughly characterized across three datasets, three seed replicates, and two training horizons (20 and 50 epochs) in the main paper. Including DeiT-S in the supplementary experiments would add six additional training runs without meaningfully extending the analysis beyond the BN–LN axis already captured by the ResNet-50/ConvNeXt-T pair.

P Corruption Robustness

We evaluate corruption robustness on CIFAR-100-C Hendrycks and Dietterich [2019], which applies 19 corruption types at 5 severity levels to the CIFAR-100 test set. We report mean corruption error (mCE), normalized so that same-architecture, same-normalization AdamW = 100%; lower is better.

Table 20: Corruption robustness (mCE%, ↓ is better). Baseline: same-architecture, same-normalization AdamW = 100%.

Architecture	Optimizer	Mean Corr. Acc (%)	mCE (%)
ResNet-50 (BN)	SOAP	22.74	**98.6**
	Shampoo	21.94	99.8
	AdamW	22.40	100.0
	Muon	21.59	100.5
ResNet-50 (LN)	Muon	27.44	**92.6**
	AdamW	22.93	100.0
ConvNeXt-T	Muon	31.31	**90.5**
	SOAP	26.23	99.1
	AdamW	26.23	100.0
	Shampoo	23.55	103.9
DeiT-S	Muon	28.30	**89.2**
	SOAP	21.47	98.3
	AdamW	20.89	100.0
	Shampoo	18.77	102.6
ResNet-50 (GN32)	Muon	28.75	**90.0**
	SOAP	27.88	91.3
	Shampoo	26.96	92.6
	AdamW	22.58	100.0
	SGD	14.36	112.9

Muon achieves the lowest mCE on three of four architecture configurations (ConvNeXt-T: 90.5%, DeiT-S: 89.2%, ResNet-50-LN: 92.6%), with the robustness advantage most pronounced on layer-normalized architectures. On ResNet-50 with BatchNorm, differences among optimizers are marginal (mCE 98.6–100.5%), consistent with BatchNorm's known loss-landscape smoothing effect. Shampoo underperforms AdamW on ConvNeXt-T (mCE 103.9%) and DeiT-S (102.6%), while performing near-baseline on ResNet-50-BN (99.8%), indicating that full-matrix Kronecker preconditioning does not translate to corruption robustness. Category-level analysis reveals that Muon's advantage concentrates in weather and digital corruptions (brightness, saturate, snow, frost, spatter), while noise corruptions (Gaussian, shot, impulse) show smaller inter-optimizer differences.

Dose-response pattern in corruption robustness. The GN(32) configuration extends the dose-response pattern observed in clean accuracy (Section **??**) to corruption robustness. Under GN(32), all three spectral optimizers improve mean corruption accuracy by 5–7 pp relative to their BN counterparts (Muon: $+7.2$ pp, SOAP: $+5.1$ pp, Shampoo: $+5.0$ pp), while AdamW remains flat ($+0.2$ pp). SGD under GN(32) degrades to 14.36% mean corruption accuracy (mCE 112.9%), an 8.2 pp drop below GN(32) AdamW, mirroring the clean-accuracy collapse observed in the normalization-swap experiment. This three-way divergence—spectral methods improve, AdamW holds steady, SGD degrades—constitutes a dose-response pattern: as normalization smoothing is reduced from BN to GN(32), the optimizer's capacity for spectral preconditioning determines whether it can compensate for the lost implicit conditioning, in corruption robustness as in clean accuracy.